Decent Coaching Classes

X Semi

Mind Your Math VIII

Time: 1 Hour **Marks: 30**

Q.1 (A) Four alternative answers are given for each sub-question. Choose the correct one. (4)

(1) Which of the following is a linear equation in two variables?

(A) $x^2 + y = 5$ (B) $xy = 4$ (C) $3x + 2y = 7$ (D) $x + 1/y = 2$

(2) The value of the discriminant of the equation $2x^2 + 5x + 3 = 0$ is

(A) 49 (B) −1 (C) 0 (D) 1

(3) The common difference of the A.P. 12, 9, 6, 3, … is

(A) 3 (B) −3 (C) 12 (D) 9

(4) For the equations $2x + y = 5$ and $x - y = 1$, the value of D is

(A) −3 (B) 3 (C) −1 (D) 1

Q.1 (B) Solve the following sub-questions. (4)

(1) Write the next term of the A.P. 3, 7, 11, …

(2) Write the equation $3x^2 = 5x - 2$ in the standard form $ax^2 + bx + c = 0$.

(3) Write the equation $3x = 4y - 12$ in the general form $ax + by + c = 0$.

(4) In an A.P., $a = 7$ and $d = 2$. Find t_3.

Q.2 (A) Complete any two of the following activities. (4)

(1) Solve the simultaneous equations by Cramer's method: $3x + y = 5$; $x - y = 3$

$D = a_1b_2 - a_2b_1 = 3 \times (-1) - 1 \times$ ____ = ____

$D_x = c_1b_2 - c_2b_1 = 5 \times (-1) - 3 \times$ ____ = ____

$D_y = a_1c_2 - a_2c_1 = 3 \times$ ____ $- 1 \times 5 =$ ____

$x = D_x / D =$ ____ , $y = D_y / D =$ ____

(2) Solve by factorisation: $x^2 - 7x + 10 = 0$

$x^2 - 5x -$ ____ $+ 10 = 0$

$x(x - 5) - 2($____$) = 0$

$(x - 5)($____$) = 0 \quad \therefore x =$ ____ or $x =$ ____

(3) Find t_{12} of the A.P. 2, 5, 8, 11, …

$a = 2$, $d = 5 - 2 =$ ____, $t_n = a + (n - 1)d$

$\therefore t_{12} = 2 + (12 - 1) \times$ ____ $= 2 +$ ____ = ____

Q.2 (B) Solve any two of the following sub-questions. (4)

(1) Solve: $2x + y = 11$; $x - y = 1$

(2) Solve: $x^2 - 9x + 20 = 0$

(3) Find the 15th term of the A.P. 4, 7, 10, …

Q.3 (A) Complete any one of the following activities. (3)

(1) Solve $2x^2 - 5x + 3 = 0$ using the formula.

Comparing with $ax^2 + bx + c = 0$: $a = 2$, $b =$ ____, $c =$ ____

$b^2 - 4ac =$ ____ $-$ ____ = ____

$x = (-b \pm \sqrt{(b^2 - 4ac)}) / 2a = ($____ $\pm$ ____$) / 4$

$\therefore x =$ ____ or $x =$ ____

(2) Find the sum of the first 20 terms of the A.P. 5, 9, 13, …

$a = 5$, $d =$ ____, $n = 20$. $S_n = n/2\,[2a + (n - 1)d]$

$\therefore S_{20} = 20/2\,[2 \times$ ____ $+ 19 \times$ ____$] =$ ____ $\times$ [____] = ____

Q.3 (B) Solve any two of the following sub-questions. **(6)**

(1) Solve by Cramer's method: $2x + 3y = 13$; $3x - y = 3$

(2) Find the roots of the equation $2x^2 - 7x + 3 = 0$ by factorisation.

(3) How many terms are there in the A.P. 7, 11, 15, …, 99?

Q.4 Solve any one of the following sub-questions. **(5)**

(1) The sum of the digits of a two-digit number is 9. The number obtained by interchanging the digits is 27 more than the original number. Find the original number.

(2) The product of two consecutive positive integers is 132. Find the integers.

(3) In an A.P., the first term is 3 and the common difference is 4. Find t_{25} and S_{25}.